Supplementary Information: Publication-Level Generalization in Perovskite Solar Cell Prediction: DOI-Grouped Evaluation with Literature-Derived Recipe Features

Tier naming. Throughout the main paper and this document, $\mathcal{M}_1$ = PDP-derived device, processing and composition features; $\mathcal{M}_2 = \mathcal{M}_1$ + curated material-property descriptors; $\mathcal{M}_3 = \mathcal{M}_2$ + the 36 LLM-derived recipe flags. An earlier, superseded 44-predictor literature representation (the "earlier literature block") is *not* $\mathcal{M}_3$; where its results are shown they are labelled "$\mathcal{M}_2$ + earlier lit. block". In the underlying run files and result tables of the repository this block carries the tier name `t3`; this document relabels it and never mixes its numbers with the recipe-flag results.

Table S1. Predictors per tier

Column sets shown for a preprocessor fitted on the full filtered data (documentation); in the experiments the `_na` indicators and the >80% missing filter are re-derived from each training partition. Tiers are nested: $\mathcal{M}_1 \subset \mathcal{M}_2 \subset \mathcal{M}_3$ (77, 179 and 215 predictor columns in the common-split models).

t1: 77 predictors (77 added relative to the previous tier)

process (21): ETL_deposition_method, HTL_deposition_method, Perovskite_deposition_method, Perovskite_thickness_nm, ETL_thickness_nm, HTL_thickness_nm, Backcontact_thickness_nm, Perov_anneal_temp_C, Perov_anneal_time_min, Perov_deposition_ Perovskite_dimension_list_of_layers, Perov_quenching_crystallisation, Perovskite_dimension_3D_with_2D_capping_layer, Perovskite_deposition_solvent_annealing, Perovskite_thickness_nm_na, ETL_thickness_nm_na, HTL_thickness_nm_na, Backcontact_thickness_nm_na, Perov_anneal_temp_C_na, Perov_anneal_time_min_na, Perovskite_dimension_list_of_layers_na

device (9): Cell_architecture, Perovskite_band_gap_measured, is_flexible_x, is_semitransparent, Encapsulation, Add_lay_front, Perovskite_pl_max_nm, Perovskite_band_gap_measured_na, Perovskite_pl_max_nm_na

composition (47): num_a_ions, num_b_ions, num_c_ions, total_ion_count, a_coeff_sum, b_coeff_sum, c_coeff_sum, a_FA_fraction, a_MA_fraction, a_Cs_fraction, a_Rb_fraction, a_organic_fraction, a_other_fraction, b_Pb_fraction, b_Sn_fraction, b_Bi_fraction, b_other_fraction, halide_I_fraction, halide_Br_fraction, halide_Cl_fraction, halide_F_fraction halide_other_fraction, is_inorganic, is_leadfree, is_3D, is_2D, is_2D3D_mixture, Perovskite_composition_perovskite_inspired_ Perovskite_composition_perovskite_ABC3_structure, a_coeff_sum_na, b_coeff_sum_na, c_coeff_sum_na, a_FA_fraction_na, a_MA_fraction_na, a_Cs_fraction_na, a_Rb_fraction_na, a_organic_fraction_na, a_other_fraction_na, b_Pb_fraction_na, b_Sn_fraction_na, b_Bi_fraction_na, b_other_fraction_na, halide_I_fraction_na, halide_Br_fraction_na, halide_Cl_fraction_na halide_F_fraction_na, halide_other_fraction_na

t2: 179 predictors (102 added relative to the previous tier)

material (102): ETL_band_gap, ETL_LUMO, ETL_HOMO, ETL_electron_mobility, ETL_hole_mobility, ETL_dielectric_constant, ETL_min_atomic_mass, ETL_max_atomic_mass, ETL_avg_atomic_mass, ETL_min_radius, ETL_max_radius, ETL_avg_radius, ETL_en_min, ETL_en_max, ETL_en_avg, HTL_band_gap, HTL_LUMO, HTL_HOMO, HTL_electron_mobility, HTL_hole_mobility, HTL_dielectric_constant, HTL_min_atomic_mass, HTL_max_atomic_mass, HTL_avg_atomic_mass, HTL_min_radius, HTL_max_radius, HTL_avg_radius, HTL_en_min, HTL_en_max, HTL_en_avg, Perovskite_band_gap, Perovskite_LUMO, Perovskite_HOMO, Perovskite_electro Perovskite_hole_mobility, Perovskite_dielectric_constant, Perovskite_min_atomic_mass, Perovskite_max_atomic_mass, Perovskite_avg_atomic_mass, Perovskite_min_radius, Perovskite_max_radius, Perovskite_avg_radius, Perovskite_en_min, Perovskite_en_max, Perovskite_en_avg, perovskite_tolerance_factor, perovskite_octahedral_factor, perovskite_cation_radius_ perovskite_tau_descriptor, HTL_hole_barrier, ETL_electron_barrier, ETL_band_gap_na, ETL_LUMO_na, ETL_HOMO_na, ETL_electron_mobility_na, ETL_hole_mobility_na, ETL_dielectric_constant_na, ETL_min_atomic_mass_na, ETL_max_atomic_mass_na ETL_avg_atomic_mass_na, ETL_min_radius_na, ETL_max_radius_na, ETL_avg_radius_na, ETL_en_min_na, ETL_en_max_na, ETL_en_avg_na, HTL_band_gap_na, HTL_LUMO_na, HTL_HOMO_na, HTL_electron_mobility_na, HTL_hole_mobility_na, HTL_dielectric_con HTL_min_atomic_mass_na, HTL_max_atomic_mass_na, HTL_avg_atomic_mass_na, HTL_min_radius_na, HTL_max_radius_na, HTL_avg_radius_na, HTL_en_min_na, HTL_en_max_na, HTL_en_avg_na, Perovskite_band_gap_na, Perovskite_LUMO_na, Perovskite_HOMO_ Perovskite_electron_mobility_na, Perovskite_hole_mobility_na, Perovskite_dielectric_constant_na, Perovskite_min_atomic_ma

Perovskite_max_atomic_mass_na, Perovskite_avg_atomic_mass_na, Perovskite_min_radius_na, Perovskite_max_radius_na, Perovskite_avg_radius_na, Perovskite_en_min_na, Perovskite_en_max_na, Perovskite_en_avg_na, perovskite_tolerance_factor_na perovskite_octahedral_factor_na, perovskite_cation_radius_ratio_na, perovskite_tau_descriptor_na, HTL_hole_barrier_na, ETL_electron_barrier_na

t3: 215 predictors (36 recipe flags added to $\mathcal{M}_2$)

Flags are paper-level, binary, and missing (native NaN, no indicator) for publications without an extraction record. Share positive among the 2,835 extracted papers in parentheses: rc_cl_source (20.5%), rc_acetate (4.7%), rc_scn (2.3%), rc_snf2 (0.7%), rc_ava (0.8%), rc_hx_acid (7.0%), rc_alkali (2.4%), rc_cs (12.0%), rc_pea_family (60.8%), rc_lewis_base (34.7%), rc_water_add (4.1%), rc_polymer (17.7%), rc_carbon (28.4%), rc_dio (0.7%), rc_htl_tbp (24.5%), rc_htl_litfsi (23.5%), rc_htl_fk (4.9%), rc_htl_f4tcnq (0.8%), rc_htl_cu (4.5%), rc_htl_borate (0.3%), rc_etl_ticl4 (5.6%), rc_etl_dopant (6.2%), rc_undoped (9.1%), rc_interface_pass (15.7%), rc_antisolvent (6.4%), rc_vacuum (17.6%), rc_two_step (3.9%), rc_hot_cast (0.0%), rc_solvent_anneal (1.0%), rc_flexible (4.3%), rc_tandem (3.4%), rc_lead_free (4.0%), rc_inorganic (7.5%), rc_mesoscopic (22.3%), rc_n2_box (4.0%), rc_ambient (10.4%).

Earlier literature block (44 predictors; not part of $\mathcal{M}_3$)

literature (44): has_BCP, has_LiF, has_PEAI, has_SAM, has_C60, has_PCBM, has_Spiro, has_PEDOT, has_NiOx, has_SnO2, has_TiO2, has_ZnO, has_PTAA, has_fullerene, has_passivation_layer, is_flexible_y, is_encapsulated, has_stability_measurement, used_antisolvent, used_glovebox, used_vacuum_evaporation, json_numeric_values_found, has_BCP_na, has_LiF_na, has_PEAI_na, has_SAM_na, has_C60_na, has_PCBM_na, has_Spiro_na, has_PEDOT_na, has_NiOx_na, has_SnO2_na, has_TiO2_na, has_ZnO_na, has_PTAA_na, has_fullerene_na, has_passivation_layer_na, is_flexible_y_na, is_encapsulated_na, has_stability_measurement_na used_antisolvent_na, used_glovebox_na, used_vacuum_evaporation_na, json_numeric_values_found_na

Table S2. Per-split results of the matched CatBoost five-holdout experiment (DOI-grouped; tier key t3 in the run files = earlier literature block, not the final $\mathcal{M}_3$; final $\mathcal{M}_3$ results: Tables S14–S17)

Subset	Tier	Split	n_{test}	R^2	MAE	RMSE
nip	t1	1	5476	0.3524	3.101	3.937
nip	t1	2	5376	0.3780	3.067	3.928
nip	t1	3	5688	0.3845	3.068	3.936
nip	t1	4	5785	0.3923	2.983	3.814
nip	t1	5	5736	0.3652	3.091	3.976
nip	t2	1	5476	0.4006	2.931	3.788
nip	t2	2	5376	0.4228	2.933	3.784
nip	t2	3	5688	0.4030	2.999	3.876
nip	t2	4	5785	0.4049	2.957	3.775
nip	t2	5	5736	0.3877	3.007	3.905
nip	t2 + earlier lit. block	1	5476	0.3939	2.960	3.809
nip	t2 + earlier lit. block	2	5376	0.4197	2.953	3.794
nip	t2 + earlier lit. block	3	5688	0.4091	2.982	3.856
nip	t2 + earlier lit. block	4	5785	0.4071	2.943	3.768
nip	t2 + earlier lit. block	5	5736	0.3963	2.981	3.877
overall	t1	1	8224	0.3707	3.082	3.914
overall	t1	2	8505	0.3345	3.101	3.958
overall	t1	3	8269	0.3604	3.136	3.945
overall	t1	4	8550	0.3990	3.036	3.859
overall	t1	5	8594	0.3767	3.051	3.862
overall	t2	1	8224	0.4145	2.943	3.775
overall	t2	2	8505	0.3712	2.985	3.847
overall	t2	3	8269	0.3787	3.067	3.888
overall	t2	4	8550	0.4420	2.892	3.719
overall	t2	5	8594	0.4326	2.864	3.684
overall	t2 + earlier lit. block	1	8224	0.4089	2.966	3.793
overall	t2 + earlier lit. block	2	8505	0.3731	2.982	3.842

Subset	Tier	Split	n_{test}	R^2	MAE	RMSE
overall	t2 + earlier lit. block	3	8269	0.3782	3.075	3.890
overall	t2 + earlier lit. block	4	8550	0.4408	2.893	3.723
overall	t2 + earlier lit. block	5	8594	0.4317	2.874	3.687
pin	t1	1	2437	0.3046	3.116	3.919
pin	t1	2	2521	0.1653	3.166	4.151
pin	t1	3	2723	0.2863	3.123	4.020
pin	t1	4	2872	0.2065	3.202	4.200
pin	t1	5	2441	0.2596	3.231	4.134
pin	t2	1	2437	0.3447	3.026	3.804
pin	t2	2	2521	0.2043	3.081	4.053
pin	t2	3	2723	0.3437	2.994	3.855
pin	t2	4	2872	0.2379	3.105	4.116
pin	t2	5	2441	0.3248	3.028	3.948
pin	t2 + earlier lit. block	1	2437	0.3592	2.980	3.762
pin	t2 + earlier lit. block	2	2521	0.1815	3.136	4.111
pin	t2 + earlier lit. block	3	2723	0.3385	3.009	3.871
pin	t2 + earlier lit. block	4	2872	0.2571	3.105	4.064
pin	t2 + earlier lit. block	5	2441	0.3315	3.026	3.928

Table S3. Split integrity (all 270 recorded fits)

DOI-grouped fits: 135; maximum DOI overlap between test and train: 0, validation and train: 0, test and validation: 0.

Records without DOI: 232 (one block). Publications (normalised DOIs): 7326.

Data SHA-256: `600efce587a1bd81c2a192e36db6534d8d78a3edfadb635dd40e685c9314ffdc`; config SHA-256: `dddb51a44e89fde8`; git HEAD at run time: `9b1d2ca3e1c7` (working tree dirty: True); CatBoost 1.2.10, scikit-learn 1.6.1, Python 3.13.5.

Table S4. Earlier literature block: coverage and constancy within publication

Feature	coverage	constant within DOI
contact_thickness_nm	0.176	1.000
etl_anneal_temp_C	0.053	1.000
etl_thickness_nm	0.101	1.000
has_BCP	0.362	1.000
has_C60	0.362	1.000
has_LiF	0.362	1.000
has_NiOx	0.362	1.000
has_PCBM	0.362	1.000
has_PEAI	0.362	1.000
has_PEDOT	0.362	1.000
has_PTAA	0.362	1.000
has_SAM	0.362	1.000
has_SnO2	0.362	1.000
has_Spiro	0.362	1.000
has_TiO2	0.362	1.000
has_ZnO	0.362	1.000
has_fullerene	0.362	1.000
has_passivation_layer	0.362	1.000
has_stability_measurement	0.362	1.000
htl_anneal_temp_C	0.037	1.000
htl_thickness_nm	0.081	1.000
is_encapsulated	0.362	1.000
is_flexible_y	0.362	1.000
json_numeric_values_found	0.362	1.000
perov_anneal_temp_C	0.070	1.000
perov_anneal_time_min	0.063	1.000
perov_concentration	0.051	1.000
perovskite_thickness_nm	0.122	1.000
spin_speed_rpm	0.100	1.000
used_antisolvent	0.362	1.000
used_glovebox	0.362	1.000
used_vacuum_evaporation	0.362	1.000

Coverage values below the median belong to sparsely reported process parameters; all other columns are non-missing exactly for the 14,896 records of publications with an extraction record.

Table S5. Earlier 44-predictor literature block: five-holdout ablation (matched CatBoost, DOI-grouped; mean ± s.d. over five splits)

This experiment evaluated the superseded literature block, not $\mathcal{M}_3$. ΔR^2 is the mean per-split difference to $\mathcal{M}_2$.

Subset	Tier	R^2	MAE	RMSE	ΔR^2 vs $\mathcal{M}_2$
Pooled	$\mathcal{M}_1$	0.368 ± 0.024	3.081 ± 0.040	3.908 ± 0.046	-0.040
Pooled	$\mathcal{M}_2$	0.408 ± 0.032	2.950 ± 0.080	3.783 ± 0.085	
Pooled	$\mathcal{M}_2$ + earlier lit. block	0.407 ± 0.031	2.958 ± 0.080	3.787 ± 0.083	-0.001
p-i-n	$\mathcal{M}_1$	0.244 ± 0.058	3.168 ± 0.049	4.085 ± 0.114	-0.047
p-i-n	$\mathcal{M}_2$	0.291 ± 0.065	3.047 ± 0.045	3.955 ± 0.131	
p-i-n	$\mathcal{M}_2$ + earlier lit. block	0.294 ± 0.074	3.051 ± 0.067	3.947 ± 0.142	+0.002
n-i-p	$\mathcal{M}_1$	0.374 ± 0.016	3.062 ± 0.047	3.918 ± 0.061	-0.029
n-i-p	$\mathcal{M}_2$	0.404 ± 0.013	2.966 ± 0.036	3.825 ± 0.060	
n-i-p	$\mathcal{M}_2$ + earlier lit. block	0.405 ± 0.010	2.964 ± 0.017	3.821 ± 0.045	+0.001

Feature-group controls (same protocol):

Subset	Variant	Feat.	R^2	MAE	RMSE
Pooled	$\mathcal{M}_1$ + earlier lit. block (no material properties)	121	0.372 ± 0.026	3.060 ± 0.053	3.895 ± 0.061
Pooled	$\mathcal{M}_2$ + earlier block minus 2 completeness flags	219	0.407 ± 0.028	2.953 ± 0.075	3.787 ± 0.078
Pooled	$\mathcal{M}_2$ + 32 noise columns	211	0.404 ± 0.029	2.970 ± 0.065	3.796 ± 0.076
p-i-n	$\mathcal{M}_1$ + earlier lit. block (no material properties)	122	0.251 ± 0.071	3.150 ± 0.058	4.065 ± 0.136
p-i-n	$\mathcal{M}_2$ + earlier block minus 2 completeness flags	220	0.292 ± 0.070	3.052 ± 0.061	3.951 ± 0.134
p-i-n	$\mathcal{M}_2$ + 32 noise columns	211	0.288 ± 0.072	3.067 ± 0.063	3.962 ± 0.144
n-i-p	$\mathcal{M}_1$ + earlier lit. block (no material properties)	120	0.378 ± 0.014	3.050 ± 0.038	3.907 ± 0.053
n-i-p	$\mathcal{M}_2$ + earlier block minus 2 completeness flags	218	0.407 ± 0.012	2.960 ± 0.024	3.816 ± 0.051
n-i-p	$\mathcal{M}_2$ + 32 noise columns	210	0.402 ± 0.009	2.976 ± 0.025	3.830 ± 0.053

Table S6. Earlier-block rows of the reference-model, model-development and stricter-protocol experiments

Reference models, pooled, five DOI-grouped splits (mean ± s.d.):

Model	Tier	R^2	MAE	RMSE
XGBoost (defaults)	$\mathcal{M}_2$ + earlier lit. block	0.377 ± 0.025	3.011 ± 0.057	3.880 ± 0.056
Random forest	$\mathcal{M}_2$ + earlier lit. block	0.352 ± 0.021	3.073 ± 0.049	3.958 ± 0.070
MLP	$\mathcal{M}_2$ + earlier lit. block	0.376 ± 0.023	3.042 ± 0.052	3.883 ± 0.056
GPR (3,000-record subsample)	$\mathcal{M}_2$ + earlier lit. block	0.352 ± 0.029	3.129 ± 0.051	3.957 ± 0.065

Model-development experiment (R^2, mean ± s.d. over five splits):

Subset	Tier	Tuned	Bag	XGB	Ens
Pooled	$\mathcal{M}_2$ + earlier lit. block	0.408 ± 0.029	0.412 ± 0.029	0.411 ± 0.024	0.418 ± 0.026
p-i-n	$\mathcal{M}_2$ + earlier lit. block	0.295 ± 0.071	0.299 ± 0.069	0.284 ± 0.077	0.300 ± 0.074
n-i-p	$\mathcal{M}_2$ + earlier lit. block	0.407 ± 0.015	0.409 ± 0.015	0.403 ± 0.015	0.413 ± 0.015

Stricter protocols, pooled CatBoost R^2 (mean ± s.d.):

Protocol	Tier	R^2	MAE
Lead-author-grouped	$\mathcal{M}_2$ + earlier lit. block	0.395 ± 0.013	2.92
Forward in time	$\mathcal{M}_2$ + earlier lit. block	0.335 ± 0.058	3.21

Figs. S1 and S2. SHAP analysis (models trained on $\mathcal{M}_2$ + the earlier 44-predictor literature block; not $\mathcal{M}_3$)

Table S7. Common-split comparison: per-seed paired ΔR^2

Reserved publications, matched default CatBoost; each seed model scored separately (the main-text values use the mean of the five seed models).

Population	Comparison	seed 0	seed 1	seed 2	seed 3	seed 4
LLM-covered	$\mathcal{M}_2 - \mathcal{M}_1$	+0.0427	+0.0431	+0.0427	+0.0414	+0.0415
LLM-covered	$\mathcal{M}_3 - \mathcal{M}_2$	-0.0012	+0.0120	+0.0059	+0.0045	+0.0092
All reserved	$\mathcal{M}_2 - \mathcal{M}_1$	+0.0323	+0.0337	+0.0328	+0.0356	+0.0301
All reserved	$\mathcal{M}_3 - \mathcal{M}_2$	+0.0012	+0.0079	+0.0017	-0.0023	+0.0034

Negative controls (default settings; ΔR^2 relative to $\mathcal{M}_2$ under the same settings): LLM-covered: five DOI-permuted flag blocks -0.0049 to +0.0015; coverage indicator only +0.0015. all reserved: five DOI-permuted flag blocks -0.0028 to +0.0007; coverage indicator only +0.0011.

Table S8. Predictor and target-leakage audit

Target: PDP field `JV_default_PCE` (renamed `final_pce`), documented in the PDP column metadata as the selected power conversion efficiency of the J–V section; identical to the PDP field in all 41,107 filtered rows (and in all 43,398 exported rows). The full per-predictor audit (predictor, tier, source field, derivation, handling, leakage risk) is in `results/revision/leakage_audit.json`.

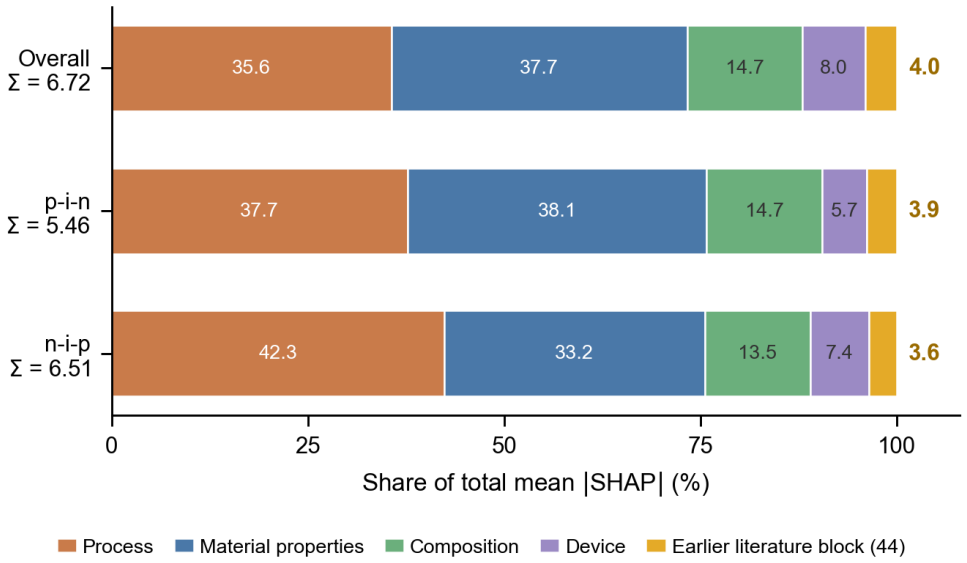


Figure 1: *

Fig. S1. Share of total feature attribution by feature group for matched CatBoost models trained on $\mathcal{M}_2$ plus the earlier 44-predictor literature block (first DOI-grouped split, 1,500 test records). Attribution is not predictive benefit, and no SHAP analysis was run for the recipe-flag model $\mathcal{M}_3$.

Tier (added predictors)	none	low	moderate
t1	72	5	0
t2	102	0	0
t3	0	36	0

Predictors of $\mathcal{M}_1$ rated low risk and retained: `Add_lay_front`, `Backcontact_thickness_nm`, `Encapsulation`, `Perovskite_band_gap_meas` `Perovskite_pl_max_nm`. These are device-context or film-characterisation fields (the absorber band gap, the photoluminescence maximum, encapsulation, an additional front layer, back-contact thickness) and not J–V, EQE, stability or stabilised-performance results. Material-property predictors ($\mathcal{M}_2$) are catalogue reference values and the recipe flags ($\mathcal{M}_3$) are paper-level identity flags; $\mathcal{M}_3$ flags are rated low because they are constant within a publication and could encode publication identity.

Checks: no predictor of the three tiers matches the outcome pattern `(^JV_|EQE|Stabilis|Stability|hysteresis|PCE|\bVoc\b|\bJsc\b|^FF$|efficien|certif|final_pce|champion|average_pce)` (hits: 0); the processed table contains five outcome-adjacent columns (`Stability_measured`, `EQE_measured`, `JV_hysteresis_index`, `Cell_area_measured_cm2`, `Stabilised_performan` all excluded in code; 118 fields of the full PDP export whose names match this pattern (J–V, EQE, stability and stabilised-performance fields among them) were never carried into the processed table. The largest absolute Spearman correlation of any numeric $\mathcal{M}_2$ predictor with the target is 0.35 (`Perovskite_avg_radius`), so no predictor is a near-copy of the target.

Table S9. Illumination sensitivity on the reserved publications (test-set exclusion, no retraining)

The full PDP export records the light spectrum and intensity. All 17 filtered records with PCE $>$ 27% (six publications) have intensity $\leq$ 0.31 (standard illumination is 100 on this scale); 12 are labelled indoor light and 5 have no spectrum label. In the filtered data 41 records have a spectrum other than AM1.5 and 249 an intensity outside 90–110. Matched default CatBoost, mean of five seed models.

Population	Test records	n	R^2 $\mathcal{M}_1$	R^2 $\mathcal{M}_2$	R^2 $\mathcal{M}_3$	MAE $\mathcal{M}_3$
LLM-covered	all	5,460	0.3177	0.3593	0.3664	2.892
LLM-covered	PCE $\leq$ 27%	5,458	0.3188	0.3602	0.3671	2.888
LLM-covered	AM1.5, 90–110 only	5,442	0.3215	0.3625	0.3696	2.879
All reserved	all	14,125	0.3712	0.4035	0.4060	2.949
All reserved	PCE $\leq$ 27%	14,117	0.3729	0.4063	0.4088	2.941
All reserved	AM1.5, 90–110 only	14,045	0.3758	0.4099	0.4124	2.930

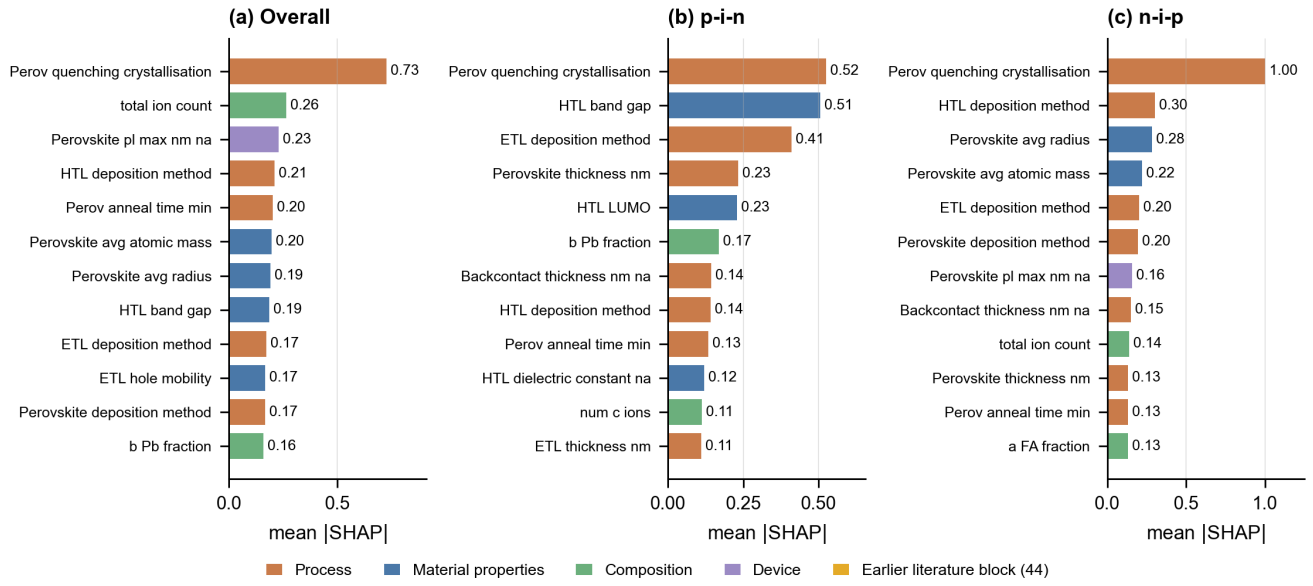


Figure 2: *

Fig. S2. Twelve highest-ranked features by mean |SHAP| (PCE percentage points) for the (a) pooled, (b) p-i-n and (c) n-i-p models of Fig. S1; `na` denotes a missingness indicator.

Table S10. Publication-weighted sensitivity metrics (reserved publications)

Each reserved record receives weight $1/n_p$, where n_p is the number of reserved records of its publication within the population; weighted $R^2 = 1 - \sum w_i e_i^2 / \sum w_i (y_i - \bar{y}_w)^2$ with the weighted mean $\bar{y}_w$; weighted MAE and RMSE are weighted means. This is equivalent to averaging the error within each publication and then across publications. Paired intervals: publication-cluster bootstrap (2000 resamples, rng seed 0).

Population	Tier	R^2 record-wtd	R^2 publication-wtd	MAE pub.-wtd	RMSE pub.-wtd
LLM-covered	$\mathcal{M}_1$	0.318	0.357	3.032	3.809
LLM-covered	$\mathcal{M}_2$	0.359	0.406	2.900	3.663
LLM-covered	$\mathcal{M}_3$	0.366	0.410	2.889	3.651
All reserved	$\mathcal{M}_1$	0.371	0.399	3.050	3.846
All reserved	$\mathcal{M}_2$	0.403	0.436	2.938	3.727
All reserved	$\mathcal{M}_3$	0.406	0.437	2.937	3.722

Population	Publication-weighted paired ΔR^2	value [95% interval]
LLM-covered	$\mathcal{M}_2 - \mathcal{M}_1$	+0.0483 [+0.0314, +0.0664]
LLM-covered	$\mathcal{M}_3 - \mathcal{M}_2$	+0.0040 [−0.0020, +0.0100]
All reserved	$\mathcal{M}_2 - \mathcal{M}_1$	+0.0366 [+0.0271, +0.0464]
All reserved	$\mathcal{M}_3 - \mathcal{M}_2$	+0.0016 [−0.0012, +0.0045]

Table S11. Extracted versus non-extracted publications (all 41,107 filtered records)

'Covered' means that the publication has one of the 2,835 extraction records. Year is the PDP `Ref_publication_date` field. Descriptive only.

Group	records	publ.	PCE mean	median	p10	p90	share ≥20%	n-i-p share
covered (extraction record exists)	15,027	2,731	12.34	12.89	5.53	18.21	2.6%	69.0%
not covered	26,080	4,596	12.47	13.08	5.03	18.56	3.6%	68.9%

Year	records	covered share
2013	223	15.7%
2014	1,468	24.5%
2015	3,392	34.0%
2016	5,946	35.2%
2017	7,556	38.6%
2018	10,128	44.8%
2019	10,122	37.5%
2020	1,502	8.9%
2021	183	0.0%
2022	406	0.0%
2024	137	0.0%

PCE bin (%)	records	covered share
(0, 8]	8,465	35.8%
(8, 12]	9,201	39.0%
(12, 16]	12,407	37.1%
(16, 20]	9,780	35.1%
(20, 40]	1,254	29.4%

Years with fewer than 100 records are omitted from the table. A two-sample Kolmogorov–Smirnov test of the PCE distributions of covered and uncovered records gives $D = 0.025$ ($p = 1.2e-05$), a small difference that is detectable only because of the sample size. Reserved partition: mean PCE 12.34 (covered) versus 12.39 (all).

Table S12. Recipe flags without the first-feature-use penalty (reserved publications, matched default CatBoost)

$\mathcal{M}_3$ refitted once with the penalty on the flag columns set to 0 instead of 0.5 (same development records, seeds, validation splits and configuration); scored on the same reserved records. The reserved publications had been examined before this run, so it is a robustness check and not independent confirmation. On development folds the penalties 0, 0.5 and 2 gave mean ΔR^2 of +0.0052, +0.0070 and +0.0038 (12, 12 and 11 of 15 folds positive).

Population	Variant	R^2	MAE	RMSE	ΔR^2 vs $\mathcal{M}_2$ [95% interval]
LLM-covered	$\mathcal{M}_2$	0.3593	2.911	3.747	
LLM-covered	$\mathcal{M}_3$, penalty 0.5 (main)	0.3664	2.892	3.726	+0.0071 [−0.0007, +0.0157]
LLM-covered	$\mathcal{M}_3$, no penalty	0.3682	2.885	3.720	+0.0089 [+0.0008, +0.0177]
All reserved	$\mathcal{M}_2$	0.4035	2.955	3.794	
All reserved	$\mathcal{M}_3$, penalty 0.5 (main)	0.4060	2.949	3.787	+0.0025 [−0.0011, +0.0062]
All reserved	$\mathcal{M}_3$, no penalty	0.4069	2.944	3.784	+0.0034 [−0.0004, +0.0072]

Table S13. Definitions of the 36 recipe flags

Each flag is 1 if the (case-insensitive) pattern occurs in the non-outcome string content of the publication's extracted material records (HTL and ETL flags: only records of that role, additive or other), and 0 otherwise. Patterns are broad lexical matches: for example `rc_pea_family` also matches "ammonium" and hence methylammonium compounds (only 104 of its 1,723 positive publications contain a spacer-specific term). Source: `src/psc/litfeat4.py`.

Flag	Group	Pattern
rc_cl_source	additive/precursor	pbcl2|macl\b|nh4cl|ch3nh3cl|methylammonium chloride|lead chloride|cscl|chloride|mapbi3-?x?cl|i3-?xclx|\bcl\b
rc_acetate	additive/precursor	acetate|pb\(ac\)|pbac2|\bmaac\b|\bpbac|ch3coo
rc_scn	additive/precursor	scn|thiocyanate
rc_snf2	additive/precursor	snf2|tin\(ii\) fluoride|tin fluoride
rc_ava	additive/precursor	5-?ava|aminovaleric|5-aminovaleric
rc_hx_acid	additive/precursor	\bhi\b|hydroiodic|hydrobromic|\bhbr\b|\bhcl\b|hypophosph|\bhpa\b|formic acid|acetic acid
rc_alkali	additive/precursor	\brbi\b|rubidium|\bki\b|potassium|\bnai\b|\bli[ib]r?\b
rc_cs	additive/precursor	\bcsi\b|cesium|caesium|\bcsbr\b|cspb
rc_pea_family	additive/precursor	\bpe(a)?i\b|phenethyl|phenylethyl|\bbai\b|butylammonium|\boai\b|octylammonium|\bgai\b|guanidin|\bnh4i|ammonium
rc_lewis_base	additive/precursor	urea|thiourea|pyridine|phosph|sulfoxide|formamide(?!.*iodide)
rc_water_add	additive/precursor	\bh2o\b|deionized|di water
rc_polymer	additive/precursor	\bpeg\b|\bpvp\b|\bpmma\b|\bps\b|polystyrene|polyethylene|\bpei\b|polymer
rc_carbon	additive/precursor	graphene|\brgo\b|cnt|nanotube|fullerene|\bc60\b|pcbm
rc_dio	additive/precursor	diiodooctane|\bdio\b|1,8-diiodo
rc_htl_tbp	HTL dopant	\btbp\b|t-bp|tert-?butyl ?pyridine|4-?tert
rc_htl_litfsi	HTL dopant	li-?tfs[ia]|lithium bis|li salt|lin\(cf3so2\)2n|litfsi

Flag	Group	Pattern
rc_htl_fk	HTL dopant	fk-?(209|102|269|211)|co\(iii\)|cobalt
rc_htl_f4tcnq	HTL dopant	f4-?tcnq|tcnq
rc_htl_cu	HTL dopant	cu\(tfsi\)2|cu\(oac\)2|cu:|copper
rc_htl_borate	HTL dopant	tpfb|borate|\bbcf\b|b\(c6f5\)3
rc_etl_ticl4	ETL treatment	ticl4
rc_etl_dopant	ETL treatment	\bnb\b|\bmg:|mg-?doped|zn-?doped|ni-?doped|\bli-doped|y-doped|nb-?doped|sn-doped|doped
rc_antisolvent	process route	anti-?solvent|quench
rc_vacuum	process route	vacuum|evaporat|co-?evaporat|vapou?r|sublim
rc_two_step	process route	two-?step|sequential|dipping|\bdip\b
rc_hot_cast	process route	hot-?cast
rc_solvent_anneal	process route	solvent anneal|dmf vapou?r|vapou?r anneal
rc_n2_box	process route	glove ?box|nitrogen|\bn2\b|argon
rc_ambient	process route	ambient|humid|\bair\b
rc_flexible	device context	\bpet\b|\bpen\b|flexible|polyimide
rc_tandem	device context	tandem|silicon|\bsi\b
rc_lead_free	device context	snI3|sni2|bismuth|\bbii3|antimony|germanium|tin-?based|lead-?free
rc_inorganic	device context	cspbi|cspbbr|cssni|inorganic
rc_mesoscopic	device context	mesoporous|mp-?tio2|scaffold|carbon electrode
rc_undoped	undoped/interface	undoped|pristine|without (additive|dopant)|dopant-free|dopant free
rc_interface_pass	undoped/interface	passivat|interlayer|surface treatment|post-?treatment|modif

Table S14. Final tier $\mathcal{M}_3$: run integrity

The $\mathcal{M}_3$ arm ($\mathcal{M}_2$ + 36 recipe flags; tier key `t3f` in `results/t3final_matched_v1`) comprises 108 fits (CatBoost and ridge; Record-level, DOI-grouped, Lead-author, Forward protocols; pooled, p-i-n and n-i-p data). Test indices equal those of the stored $\mathcal{M}_2$ arm in all 108 fits for which a stored arm exists. Maximum DOI overlap (test/train, validation/train, test/validation) in the publication-disjoint protocols: Lead-author 0/0/0, DOI-grouped 0/0/0, Forward 0/0/0. CatBoost first-feature-use penalty on the flag columns: 0.5; ridge uses training-mean imputed flags plus a coverage indicator. Data SHA-256 `600efce587a1bd81`, recipe artifact SHA-256 `3b8670b2d6182520`, git HEAD at run time `62d87bdd39e0`.

Table S15. Per-split paired $\mathcal{M}_3 - \mathcal{M}_2$ differences (pooled data)

ΔR^2 with the 95% publication-cluster bootstrap interval of each split (1,000 resamples); n test records.

Protocol	Model	Scope	Split	n	R^2 $\mathcal{M}_2$	R^2 $\mathcal{M}_3$	ΔR^2	[95% interval]
Lead-author	catboost	all	1	7,746	0.4101	0.4144	+0.0043	[−0.0046, +0.0121]
Lead-author	catboost	all	2	9,424	0.3850	0.3877	+0.0027	[−0.0048, +0.0105]
Lead-author	catboost	all	3	8,438	0.4131	0.4057	−0.0074	[−0.0131, −0.0016]
Lead-author	catboost	all	4	9,424	0.3706	0.3796	+0.0090	[+0.0009, +0.0183]
Lead-author	catboost	all	5	6,368	0.3898	0.3910	+0.0011	[−0.0058, +0.0080]
Lead-author	catboost	covered	1	2,594	0.3693	0.3871	+0.0178	[+0.0029, +0.0345]
Lead-author	catboost	covered	2	3,809	0.3597	0.3599	+0.0003	[−0.0132, +0.0141]
Lead-author	catboost	covered	3	3,191	0.3726	0.3713	−0.0013	[−0.0118, +0.0104]
Lead-author	catboost	covered	4	3,390	0.3338	0.3530	+0.0192	[+0.0022, +0.0383]
Lead-author	catboost	covered	5	2,275	0.3679	0.3685	+0.0006	[−0.0133, +0.0149]
Lead-author	ridge	all	1	7,746	0.3659	0.3661	+0.0002	[−0.0060, +0.0062]
Lead-author	ridge	all	2	9,424	0.3504	0.3525	+0.0021	[−0.0037, +0.0082]
Lead-author	ridge	all	3	8,438	0.3738	0.3699	−0.0039	[−0.0100, +0.0017]
Lead-author	ridge	all	4	9,424	0.3319	0.3331	+0.0012	[−0.0068, +0.0109]
Lead-author	ridge	all	5	6,368	0.3449	0.3456	+0.0006	[−0.0075, +0.0089]
Lead-author	ridge	covered	1	2,594	0.3011	0.3037	+0.0026	[−0.0163, +0.0231]
Lead-author	ridge	covered	2	3,809	0.3313	0.3373	+0.0060	[−0.0107, +0.0223]
Lead-author	ridge	covered	3	3,191	0.3331	0.3267	−0.0064	[−0.0213, +0.0085]
Lead-author	ridge	covered	4	3,390	0.3024	0.3041	+0.0017	[−0.0216, +0.0253]
Lead-author	ridge	covered	5	2,275	0.3298	0.3352	+0.0054	[−0.0162, +0.0259]
DOI-grouped	catboost	all	1	8,224	0.4145	0.4060	−0.0085	[−0.0179, +0.0004]
DOI-grouped	catboost	all	2	8,505	0.3712	0.3779	+0.0067	[−0.0036, +0.0165]
DOI-grouped	catboost	all	3	8,269	0.3787	0.3798	+0.0010	[−0.0089, +0.0106]
DOI-grouped	catboost	all	4	8,550	0.4420	0.4396	−0.0024	[−0.0138, +0.0074]
DOI-grouped	catboost	all	5	8,594	0.4326	0.4297	−0.0029	[−0.0134, +0.0094]
DOI-grouped	catboost	covered	1	2,996	0.4044	0.3920	−0.0124	[−0.0314, +0.0064]
DOI-grouped	catboost	covered	2	2,878	0.3398	0.3419	+0.0021	[−0.0172, +0.0216]
DOI-grouped	catboost	covered	3	3,162	0.3756	0.3753	−0.0003	[−0.0196, +0.0209]

Protocol	Model	Scope	Split	n	R^2 $\mathcal{M}_2$	R^2 $\mathcal{M}_3$	ΔR^2	[95% interval]
DOI-grouped	catboost	covered	4	2,894	0.4424	0.4525	+0.0101	[−0.0060, +0.0271]
DOI-grouped	catboost	covered	5	3,320	0.3609	0.3615	+0.0005	[−0.0258, +0.0321]
DOI-grouped	ridge	all	1	8,224	0.3700	0.3690	−0.0010	[−0.0074, +0.0054]
DOI-grouped	ridge	all	2	8,505	0.3022	0.2997	−0.0025	[−0.0090, +0.0043]
DOI-grouped	ridge	all	3	8,269	0.3416	0.3364	−0.0052	[−0.0136, +0.0020]
DOI-grouped	ridge	all	4	8,550	0.3788	0.3773	−0.0014	[−0.0076, +0.0043]
DOI-grouped	ridge	all	5	8,594	0.3757	0.3727	−0.0030	[−0.0094, +0.0035]
DOI-grouped	ridge	covered	1	2,996	0.3698	0.3665	−0.0033	[−0.0218, +0.0154]
DOI-grouped	ridge	covered	2	2,878	0.3002	0.2872	−0.0131	[−0.0286, +0.0027]
DOI-grouped	ridge	covered	3	3,162	0.3353	0.3247	−0.0107	[−0.0329, +0.0099]
DOI-grouped	ridge	covered	4	2,894	0.3979	0.3916	−0.0064	[−0.0217, +0.0094]
DOI-grouped	ridge	covered	5	3,320	0.2969	0.2910	−0.0059	[−0.0234, +0.0120]
Record-level	catboost	all	1	8,222	0.6226	0.6341	+0.0115	[+0.0071, +0.0159]
Record-level	catboost	all	2	8,222	0.6118	0.6207	+0.0089	[+0.0051, +0.0129]
Record-level	catboost	all	3	8,222	0.6245	0.6340	+0.0095	[+0.0052, +0.0136]
Record-level	catboost	all	4	8,222	0.6109	0.6229	+0.0120	[+0.0075, +0.0166]
Record-level	catboost	all	5	8,222	0.6190	0.6312	+0.0122	[+0.0080, +0.0166]
Record-level	catboost	covered	1	2,961	0.5944	0.6269	+0.0324	[+0.0220, +0.0422]
Record-level	catboost	covered	2	3,058	0.5996	0.6295	+0.0299	[+0.0216, +0.0379]
Record-level	catboost	covered	3	2,961	0.6088	0.6369	+0.0281	[+0.0185, +0.0384]
Record-level	catboost	covered	4	3,020	0.5688	0.6040	+0.0352	[+0.0238, +0.0470]
Record-level	catboost	covered	5	2,942	0.5948	0.6273	+0.0325	[+0.0221, +0.0428]
Record-level	ridge	all	1	8,222	0.4123	0.4155	+0.0033	[+0.0002, +0.0063]
Record-level	ridge	all	2	8,222	0.3946	0.3982	+0.0036	[+0.0005, +0.0067]
Record-level	ridge	all	3	8,222	0.4155	0.4195	+0.0040	[+0.0011, +0.0070]
Record-level	ridge	all	4	8,222	0.4075	0.4116	+0.0040	[+0.0011, +0.0073]
Record-level	ridge	all	5	8,222	0.4118	0.4166	+0.0047	[+0.0019, +0.0079]
Record-level	ridge	covered	1	2,961	0.3889	0.4001	+0.0111	[+0.0033, +0.0194]
Record-level	ridge	covered	2	3,058	0.3825	0.3911	+0.0086	[−0.0004, +0.0173]
Record-level	ridge	covered	3	2,961	0.4172	0.4292	+0.0120	[+0.0046, +0.0197]
Record-level	ridge	covered	4	3,020	0.3791	0.3924	+0.0133	[+0.0046, +0.0219]
Record-level	ridge	covered	5	2,942	0.3928	0.4051	+0.0123	[+0.0033, +0.0211]
Forward	catboost	all	1	6,502	0.3347	0.3361	+0.0013	[−0.0103, +0.0127]
Forward	catboost	all	2	6,322	0.3984	0.4030	+0.0046	[−0.0035, +0.0128]
Forward	catboost	all	3	6,484	0.3020	0.2895	−0.0125	[−0.0259, +0.0009]
Forward	catboost	covered	1	2,870	0.3395	0.3336	−0.0059	[−0.0279, +0.0143]
Forward	catboost	covered	2	2,556	0.4085	0.4153	+0.0068	[−0.0057, +0.0201]
Forward	catboost	covered	3	1,564	0.3432	0.3376	−0.0055	[−0.0300, +0.0202]
Forward	ridge	all	1	6,502	0.2945	0.2890	−0.0055	[−0.0155, +0.0039]
Forward	ridge	all	2	6,322	0.3690	0.3833	+0.0143	[+0.0039, +0.0239]
Forward	ridge	all	3	6,484	0.1956	0.1959	+0.0003	[−0.0066, +0.0068]
Forward	ridge	covered	1	2,870	0.2733	0.2578	−0.0155	[−0.0356, +0.0061]
Forward	ridge	covered	2	2,556	0.3811	0.3908	+0.0098	[−0.0101, +0.0306]
Forward	ridge	covered	3	1,564	0.3447	0.3481	+0.0034	[−0.0173, +0.0245]

Table S16. Paired $\mathcal{M}_3 - \mathcal{M}_2$ summary by architecture subset

Mean over splits; "pos." counts splits with positive ΔR^2; "CI> 0/< 0" counts splits whose interval lies above/below zero. Record-level p-i-n and n-i-p CatBoost fits compare default-iteration arms for both tiers (not the converged pooled fits).

Protocol	Subset	Model	Scope	n	R^2 $\mathcal{M}_2$	R^2 $\mathcal{M}_3$	ΔR^2	ΔMAE	pos.	CI> 0/< 0
Record-level	pooled	catboost	all	8,222	0.618	0.629	+0.0108	−0.039	5/5	5/0
Record-level	pooled	catboost	covered	2,988	0.593	0.625	+0.0316	−0.115	5/5	5/0
Record-level	pooled	ridge	all	8,222	0.408	0.412	+0.0039	−0.013	5/5	5/0
Record-level	pooled	ridge	covered	2,988	0.392	0.404	+0.0115	−0.038	5/5	4/0
Record-level	p-i-n	catboost	all	2,537	0.571	0.578	+0.0074	−0.018	5/5	3/0
Record-level	p-i-n	catboost	covered	927	0.524	0.546	+0.0220	−0.065	5/5	4/0
Record-level	p-i-n	ridge	all	2,537	0.385	0.390	+0.0052	−0.012	5/5	1/0
Record-level	p-i-n	ridge	covered	927	0.352	0.364	+0.0115	−0.025	5/5	1/0
Record-level	n-i-p	catboost	all	5,668	0.605	0.616	+0.0109	−0.037	5/5	5/0
Record-level	n-i-p	catboost	covered	2,080	0.585	0.623	+0.0377	−0.129	5/5	5/0
Record-level	n-i-p	ridge	all	5,668	0.435	0.440	+0.0052	−0.019	5/5	4/0
Record-level	n-i-p	ridge	covered	2,080	0.411	0.426	+0.0148	−0.052	5/5	4/0
DOI-grouped	pooled	catboost	all	8,428	0.408	0.407	−0.0012	+0.006	2/5	0/0
DOI-grouped	pooled	catboost	covered	3,050	0.385	0.385	+0.0000	+0.009	3/5	0/0
DOI-grouped	pooled	ridge	all	8,428	0.354	0.351	−0.0026	+0.006	0/5	0/0

Protocol	Subset	Model	Scope	n	R^2 $\mathcal{M}_2$	R^2 $\mathcal{M}_3$	ΔR^2	ΔMAE	pos.	CI$>0/<0$
DOI-grouped	pooled	ridge	covered	3,050	0.340	0.332	−0.0079	+0.014	0/5	0/0
DOI-grouped	p-i-n	catboost	all	2,599	0.291	0.288	−0.0032	+0.018	3/5	0/0
DOI-grouped	p-i-n	catboost	covered	912	0.269	0.267	−0.0021	+0.011	2/5	0/0
DOI-grouped	p-i-n	ridge	all	2,599	0.206	0.198	−0.0075	+0.024	1/5	0/1
DOI-grouped	p-i-n	ridge	covered	912	0.211	0.191	−0.0202	+0.059	1/5	0/1
DOI-grouped	n-i-p	catboost	all	5,612	0.404	0.403	−0.0004	+0.003	2/5	0/0
DOI-grouped	n-i-p	catboost	covered	2,052	0.394	0.396	+0.0016	−0.007	3/5	0/0
DOI-grouped	n-i-p	ridge	all	5,612	0.367	0.365	−0.0022	+0.008	1/5	0/0
DOI-grouped	n-i-p	ridge	covered	2,052	0.365	0.365	+0.0001	−0.002	2/5	0/0
Lead-author	pooled	catboost	all	8,280	0.394	0.396	+0.0019	−0.001	4/5	1/1
Lead-author	pooled	catboost	covered	3,052	0.361	0.368	+0.0073	−0.015	4/5	2/0
Lead-author	pooled	ridge	all	8,280	0.353	0.353	+0.0000	−0.002	4/5	0/0
Lead-author	pooled	ridge	covered	3,052	0.320	0.321	+0.0019	−0.011	4/5	0/0
Lead-author	p-i-n	catboost	all	2,191	0.321	0.318	−0.0026	−0.000	3/5	1/1
Lead-author	p-i-n	catboost	covered	780	0.329	0.328	−0.0002	−0.008	3/5	0/0
Lead-author	p-i-n	ridge	all	2,191	0.272	0.275	+0.0035	−0.006	3/5	1/0
Lead-author	p-i-n	ridge	covered	780	0.297	0.292	−0.0050	+0.018	2/5	1/1
Lead-author	n-i-p	catboost	all	5,835	0.401	0.402	+0.0005	+0.001	3/5	0/0
Lead-author	n-i-p	catboost	covered	2,190	0.387	0.398	+0.0101	−0.011	3/5	1/0
Lead-author	n-i-p	ridge	all	5,835	0.369	0.369	−0.0006	+0.007	2/5	0/0
Lead-author	n-i-p	ridge	covered	2,190	0.366	0.364	−0.0015	+0.015	1/5	1/0
Forward	pooled	catboost	all	6,436	0.345	0.343	−0.0022	−0.008	2/3	0/0
Forward	pooled	catboost	covered	2,330	0.364	0.362	−0.0016	−0.006	1/3	0/0
Forward	pooled	ridge	all	6,436	0.286	0.289	+0.0031	−0.016	2/3	1/0
Forward	pooled	ridge	covered	2,330	0.333	0.332	−0.0008	−0.013	2/3	0/0
Forward	p-i-n	catboost	all	1,986	0.256	0.237	−0.0192	+0.052	0/3	0/1
Forward	p-i-n	catboost	covered	681	0.289	0.271	−0.0180	+0.061	0/3	0/0
Forward	p-i-n	ridge	all	1,986	0.236	0.231	−0.0056	+0.010	1/3	0/0
Forward	p-i-n	ridge	covered	681	0.232	0.227	−0.0052	−0.003	1/3	0/0
Forward	n-i-p	catboost	all	4,413	0.365	0.365	−0.0006	+0.007	1/3	0/0
Forward	n-i-p	catboost	covered	1,655	0.378	0.384	+0.0064	−0.016	2/3	0/0
Forward	n-i-p	ridge	all	4,413	0.325	0.321	−0.0032	+0.002	1/3	0/0
Forward	n-i-p	ridge	covered	1,655	0.352	0.349	−0.0037	−0.015	1/3	0/0

Table S17. SHAP analysis of the final tier $\mathcal{M}_3$ (first DOI-grouped split)

Matched CatBoost with the first-feature-use penalty on the recipe flags; `TreeExplainer` on 1,500 held-out records; mean |SHAP| in PCE percentage points. Attributions describe the fitted model only. The earlier SHAP analysis (Figs. S1 and S2) used the superseded 44-predictor block.

Subset	process	material	composition	device	recipe flags
pooled	35.5%	37.0%	14.2%	7.7%	5.6%
p-i-n	37.1%	38.9%	14.7%	5.0%	4.4%
n-i-p	40.9%	31.5%	14.8%	7.3%	5.5%

Twelve highest-ranked features and the five highest-ranked recipe flags (pooled model):

Rank	Feature	mean |SHAP|	Group
1	Perov_quenching_crystallisation	0.770	process
2	total_ion_count	0.285	composition
3	Perovskite_pl_max_nm_na	0.231	device
4	HTL_deposition_method	0.213	process
5	Perovskite_avg_atomic_mass	0.208	material
6	Perov_anneal_time_min	0.189	process
7	HTL_LUMO	0.186	material
8	Perovskite_deposition_method	0.183	process
9	Perovskite_avg_radius	0.171	material
10	HTL_band_gap	0.170	material
11	ETL_deposition_method	0.167	process
12	ETL_hole_mobility	0.165	material
35	rc_interface_pass	0.052	recipe
38	rc_vacuum	0.041	recipe
58	rc_htl_tbp	0.030	recipe
62	rc_etl_dopant	0.024	recipe
63	rc_carbon	0.024	recipe

Material-descriptor implementation notes

Perovskite geometry descriptors. Fixed effective radii (pm): A-site FA 253, MA 217, Cs 188, Rb 172; B-site Pb 119, Sn 110, Bi 103; X-site I 220, Br 196, Cl 181, F 133. Site radius = fraction-weighted mean over the listed species of that site, divided by the sum of their fractions; $\mu = r_B/r_X$, $t = (r_A + r_X)/[\sqrt{2}(r_B + r_X)]$, r_A/r_X, $\tau = r_X/r_B - (1 - (r_A/r_B)\ln(r_A/r_B))$ (script `opus/evaluate_tolerance_factor.py`; the stored octahedral and tolerance factors were reproduced to floating-point precision from the ion-fraction columns). **Layer averages.** `*_avg_radius`, `*_avg_atomic_mass` and `*_en_avg` are atom-count-weighted means of covalent radii (pm), atomic masses and Pauling electronegativities over the elements of a hand-entered formula (script `opus/fill_atomic_data.py`); `min`/`max` are the extremes over the elements. For $MAPbI_3$ ($C_1H_6N_1Pb_1I_3$) the average radius is $(77 + 6 \cdot 31 + 71 + 146 + 3 \cdot 139)/12 = 74.75$ pm, matching the stored value.

Extraction system prompt (verbatim from the recovered script)

```
You are a materials-science data-extraction assistant.

Given a Markdown document converted from a published paper, extract every
material reported with its measured or computed properties.

Return ONLY a single JSON object (no markdown fences, no commentary) with this
top-level schema:

{
  "source": "DOI: <doi-from-paper>",
  "materials": [
    {
      "formula": "<chemical formula>",
      "common_name": "<human-readable name>",
      "type": "<e.g. perovskite, ETL, HTL, metal contact>",
      ... <all properties you can find: band_gap, PV_performance, deposition_method, etc.>
    }
  ]
}

Rules:
- Always include units and measurement method when available.
- Include error bars / standard deviations if reported.
- For PV performance, separate champion vs average and HTM-free vs with-HTM.
- If the DOI is embedded in the document, use it for the "source" field.
- Do NOT invent data. Only extract what is explicitly stated.
- Return raw JSON only - no surrounding text, no code fences.
```

Model-development grid

CatBoost depth/L2 pairs: (6,3), (8,3), (10,3), (8,10), (10,10), (8,1); learning rate 0.05; up to 3,000 iterations; early stopping 150 rounds on the validation set; three seeds (42, 7, 123) for the selected configuration; XGBoost: depth 8, learning rate 0.05, 1,500 trees, subsample 0.8, colsample 0.7, $\lambda = 3$, early stopping 100 rounds.